Corrosion Science 205 (2022) 110464

Contents lists available at ScienceDirect

Corrosion Science

journal homepage: www.elsevier.com/locate/corsci

Oxidation behaviour of eutectic refractory high-entropy alloys at 800–1000 °C

N. Yurchenko *, E. Panina, S. Zherebtsov, N. Stepanov

Laboratory of Bulk Nanostructured Materials, Belgorod National Research University, Belgorod 308015, Russia

ARTICLE INFO

Keywords:
Refractory high-entropy alloys
Oxidation
Eutectic structure
High temperatures
Oxidation kinetics

ABSTRACT

Herein, we investigated the oxidation behaviour of B2/C14 Laves phase hypoeutectic $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$, eutectic $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$, and hypereutectic $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ (at%) refractory high-entropy alloys (RHEAs) at 800–1000 °C. An increase in the Al/Ti ratio reduced the mass gain, but induced an oxide scale spallation. Before spallation, the studied alloys demonstrated better oxidation resistance than the current Al-Cr-Nb-Ti-(V)-Zr RHEAs at 800 °C. An analysis of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy with near-parabolic oxidation kinetics revealed the formation of an external oxide layer with a dominant TiO_2 and an internal reaction zone comprised O-rich products and TiN. The oxidation mechanisms and suggestions for improving the oxidation/spallation resistance were discussed.

1. Introduction

The growing demand for enhanced economic efficiency of aircraft engines requires new materials with advanced performance. Particular interest exists in substituting Ni-based superalloys to increase gas turbine inlet temperature [1]. The recently introduced refractory high-entropy alloys (RHEAs) offer enormous opportunities to solve this issue [2,3]. For instance, some RHEAs had already demonstrated good strength or oxidation resistance at $T \geq 1200$ °C [4–11].

Meanwhile, light alloys capable of operating at 600–1000 °C are also highly desirable for improving the thrust-to-weight ratio [12]. RHEAs based on the Al-Cr-Nb-Ti-V-Zr system with densities < 6.5 g/cm^3 have great potential to fill this temperature interval [13–25]. Systematic investigations revealed numerous Al-Cr-Nb-Ti-V-Zr RHEAs with high specific strength at $T \leq 1000$ °C, outperforming Ti-based intermetallics and Ni-based superalloys [13–16,22]. However, compared to mechanical properties, the oxidation behaviour of these alloys is less studied. V-containing RHEAs had poor oxidation resistance at $T \geq 700$ °C due to the formation of volatile oxides, alleviating oxygen penetration through a porous oxide layer [26–28]. Moreover, a recent attempt to protect an AlTiCrVZr RHEA via micro-arc oxidation coatings was unsuccessful [29]. V-free RHEAs showed better oxidation resistance, but, even with the relatively high Al and/or Cr content, none of the alloys could form "classical", continuous Al_2O_3/Cr_2O_3 oxide layers. Reasonable protection against oxygen attack, often associated with internal oxidation, was ensured by multiple oxides, including complex ones such as $CrNbO_4$, $AlNbO_4$, and Ti_2ZrO_6 [26,30–32].

Unfortunately, an endeavour to connect the mass gain with the chemical composition of the Al-Cr-Nb-Ti-Zr RHEAs to design oxidation-resistant alloys is a challenge. Scarce experimental data, collected in Table 1, suggest that simultaneous Al and Cr presence in equiatomic alloys provides better oxidation resistance than sole Al or Cr additions. However, in the non-equiatomic Al-Cr-Nb-Ti-Zr RHEAs, there is no direct correlation between the mass gain and the concentration of elements. For instance, an $Al_{20}Cr_{25}Nb_{20}Ti_{20}Zr_{15}$ (at%) RHEA reported by Gawel et al. [33] gained 21 mg/cm^2 after 24 h at 800 °C, which was ~2.5 times higher than that in an equiatomic $Al_{20}Cr_{20}Nb_{20}Ti_{20}Zr_{20}$ alloy [31]. In turn, Waseem et al. claimed a positive effect of the reduced Zr content on oxidation resistance at 1000 °C by presenting an $Al_{12.5}Cr_{25}Nb_{25}Ti_{25}Zr_{12.5}$ RHEA [32]. Such an elemental redistribution resulted in significant but not fully clarified enhancement of oxidation resistance compared to their former $Al_{10}Cr_{30}Nb_{20}Ti_{20}Zr_{20}$ alloy [34] (Table 1). The benefit of the newly developed $Al_{12.5}Cr_{25}Nb_{25}Ti_{25}Zr_{12.5}$ alloy over previous $Al_{20}Cr_{20}Nb_{20}Ti_{20}Zr_{20}$ and $Al_{30}Cr_{10}Nb_{20}Ti_{20}Zr_{20}$ alloys after oxidation at 1000 °C for 20 h was negligible but became prominent at a longer exposure (50 h) (Table 1).

The effect of (micro)structure on the oxidation behaviour of the Al-Cr-Nb-Ti-Zr RHEAs is also not well-established. Butler et al. showed that a Laves phase could impede, to some extent, the oxidation front in a $Cr_{25}Nb_{25}Ti_{25}Zr_{25}$ alloy in contrast to a bcc phase [26]. Meantime, the

* Corresponding author.
E-mail address: yurchenko_nikita@bsu.edu.ru (N. Yurchenko).

https://doi.org/10.1016/j.corsci.2022.110464
Received 14 February 2022; Received in revised form 16 June 2022; Accepted 24 June 2022
Available online 27 June 2022
0010-938X/© 2022 Elsevier Ltd. All rights reserved.

Table 1
Available literature data [26,30–34] on the mass gain of different Al-Cr-Nb-Ti-Zr RHEAs at 800 and 1000 °C.

Alloy	Mass gain, mg/cm^2			Ref.
	800 °C, 24 h	1000 °C, 20 h	1000 °C, 50 h	
$Cr_{25}Nb_{25}Ti_{25}Zr_{25}$	–	59	84	[26]
$Al_{25}Nb_{25}Ti_{25}Zr_{25}$	3.5[a]	39[a]	63.7[a]	[30]
$Al_{20}Cr_{20}Nb_{20}Ti_{20}Zr_{20}$	8.3	22	61	[31]
$Al_{12.5}Cr_{25}Nb_{25}Ti_{25}Zr_{12.5}$	–	17	~24	[32]
$Al_{20}Cr_{25}Nb_{20}Ti_{20}Zr_{15}$	21	–	–	[33]
$Al_{20}Cr_{25}Nb_{19}Ti_{20}Zr_{15}Y_1$	9	–	–	[33]
$Al_{10}Cr_{30}Nb_{20}Ti_{20}Zr_{20}$	–	39	–	[34]
$Al_{30}Cr_{10}Nb_{20}Ti_{20}Zr_{20}$	–	20	–	[34]

[a] calculated values

most oxidation-resistant $Al_{12.5}Cr_{25}Nb_{25}Ti_{25}Zr_{12.5}$ alloy comprised 60 % of the Laves and 40 % of the bcc phases [32]. Previously, the Laves phase was shown to improve the oxidation resistance of TiAl alloys [35–38]. Thus, it would be interesting to analyse the effect of the Laves phase on the oxidation kinetics of Al-Cr-Nb-Ti-Zr RHEAs more thoroughly. Recent studies [21,39] revealed that various eutectic bcc (B2)/Laves phase microstructures were formed in such alloys, facilitating the systematic investigations. In the current work, we aimed to evaluate the oxidation behaviour of $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$, $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$, and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ RHEAs with hypoeutectic, eutectic, and hypereutectic B2/C14 Laves phase structures at 800–1000 °C.

2. Materials and methods

The ingots of the alloys with nominal compositions of $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$, $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$, and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ (at%) were produced by vacuum arc melting of pure (≥ 99.9 wt%) elements.

Specimens for oxidation tests with nominal dimensions of 6 × 4 × 4 mm^3 were cut using an electric discharge machine from the as-cast ingots. Surfaces of the specimens were mechanically gritted, polished, and ultrasonically cleaned in isopropanol. Oxidation tests were carried out in a Nabertherm furnace under the static laboratory air at 800, 900, and 1000 °C for 100 h or to the fracture of the specimens. During the testing process, samples were placed on alumina plates, put into the preheated furnace, and removed after 1, 5, 10, 25, 50, and 100 h of the oxidation test and cooled in the laboratory air for mass measurements. The mass of the specimens before and after oxidation was measured using an HR-200 analytical balance with an accuracy of 0.1 mg to characterise the oxidation kinetics. Only bulk samples were weighed after oxidation, while spalled oxide scales or fragmented parts of the specimens, which appeared during cooling, were not included in the overall mass measurements.

Phase identification of the oxides in the scales was performed by X-ray diffraction (XRD) using a RIGAKU diffractometer with Cu-Kα radiation with the wavelength of 1.5406 Å; scanning rate and interval were 3°/min and 20–80°, respectively. Cross-sectional morphologies of the oxidised specimens were examined by a scanning electron microscope (SEM) FEI Quanta 600 FEG equipped with back-scattered electron (BSE) and energy-dispersive X-ray spectroscopy (EDS) detectors. Samples for SEM investigations were mounted in Buehler KonductoMet via Buehler SimpliMet 4000 Mounting System to preserve the integrity of the oxide scale; note that only the oxidised $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy could be mounted, while the oxidised $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$ and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys fractured completely when handling. The mounted specimens were polished using Buehler VibroMet 2 Vibratory Polisher and Buehler MasterMet 2 Suspension to mirror surfaces.

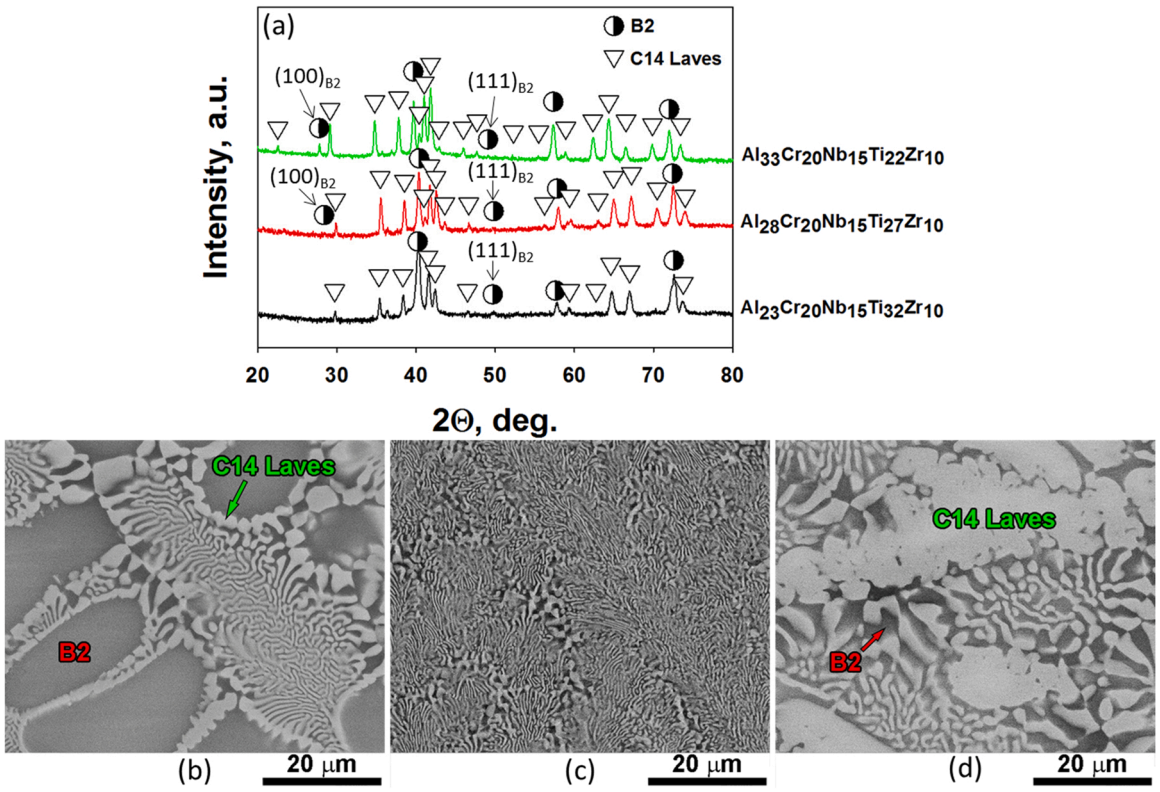


Fig. 1. The initial structure of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ (a, b), $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$ (a, c), and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ (a, d) alloys: (a) – XRD patterns with denoted Bragg's peaks of the B2 and C14 Laves phases; (b-d) – SEM-BSE images, showing the hypoeutectic (b), eutectic (c), and hypereutectic (d) B2/C14 Laves phase microstructure.

Table 2
Chemical compositions of structural constituents of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$, $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$, and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys.

Element, at.%	Al	Cr	Nb	Ti	Zr	Volume fraction, %
$Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$						
B2 phase	20.1	15.2	18.6	39.9	6.2	52
C14 Laves phase	23.7	28.2	9.7	20.8	17.6	48
Alloy composition	21.6	20.8	15.2	32.3	10.1	–
$Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$						
B2 phase	25.4	14.6	16.2	36.8	7.0	43
C14 Laves phase	31.8	31.3	9.4	13.1	14.4	57
Alloy composition	26.6	20.5	15.3	27.4	10.2	–
$Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$						
B2 phase	29.2	14.4	17.1	33.5	5.8	34
C14 Laves phase	32.1	28.0	13.9	12.5	13.5	66
Alloy composition	30.8	20.9	15.4	22.7	10.2	–

3. Results

3.1. Initial structure

A detailed structure analysis of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$, $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$, and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys was presented elsewhere [21]. Here, a brief overview of the non-oxidised alloys is given. According to XRD patterns, all three alloys showed a dual-phase structure comprising B2 (*CsCl-prototype*; *cP2*; *Pm-3 m*) and C14 Laves phases (*$MgZn_2$-prototype*; *hP12*; *$P6_3/mmc$*) (Fig. 1(a)). The Al/Ti ratio regulated the proportion between the volume fractions of the (Nb, Ti)-rich B2 and (Al, Cr, Zr)-rich C14 Laves phases (Table 2). With an increment of the Al/Ti ratio (and, correspondingly, the C14 Laves phase content), the microstructure changed from a hypoeutectic ($Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$) to a eutectic ($Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$) and then to a hypereutectic ($Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$) one (Fig. 1(b-d)).

3.2. Oxidation behaviour

Fig. 2(a-c) collect the time-dependent mass gain data obtained during interrupted oxidation tests of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$, $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$, and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys at 800–1000 °C in the laboratory air. At 800–900 °C, the mass gain decreased pronouncedly from the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ to the $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloy (Fig. 2(a-c)). Yet, in this interval, only the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy retained a pristine form after 100 h (Fig. 2(a, b, d)). The $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$ and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys showed significant spallation (Fig. 2(d)), visible as a negative mass gain, after 25 h at 800 °C and 5–25 h at 900 °C, respectively (Fig. 2(a, b)). At 1000 °C, the $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$ and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys started to spall earlier (0–1 h) than the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy, which remained intact up to 10 h. All the alloys fractured entirely after 25 h at 1000 °C.

The mass gain curves of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy were analysed by a well-known oxidation law [40]:

$$\Delta m = kt^n \quad (1)$$

where Δm – the mass gain per unit surface area (mg/cm^2); k – the

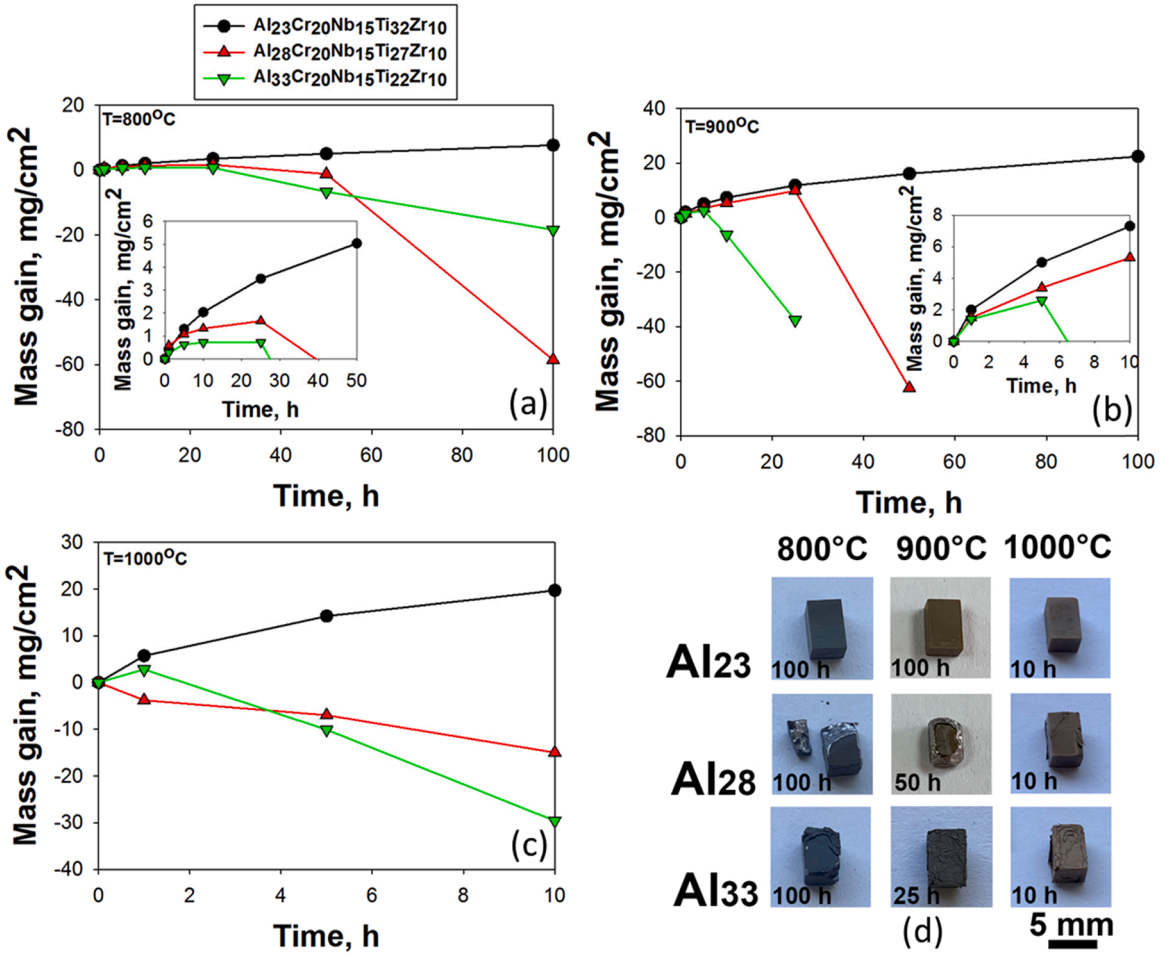


Fig. 2. Time dependences of mass gains for the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$, $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$, and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys after interrupted oxidation tests in the air at 800 °C (a), 900 °C (b), and 1000 °C (c); an overall view of the oxidised alloys after certain periods (d), showing better spallation resistance of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy compared to the $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$ and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys.

Table 3
The oxidation rate constants, k; and the time exponents, n, obtained after fitting the mass gain curves of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloys following Eq. (1).

Alloy	800 °C			900 °C			1000 °C		
	Time, h	k, mg/cm^2 h	n	Time, h	k, mg/cm^2 h	n	Time, h	k, mg/cm^2 h	n
$Al_{23}Cr_{20}Nb_{15}Ti_{33}Zr_{10}$	0–100	0.52	0.58	0–100	2.4	0.49	0–10	5.96	0.52

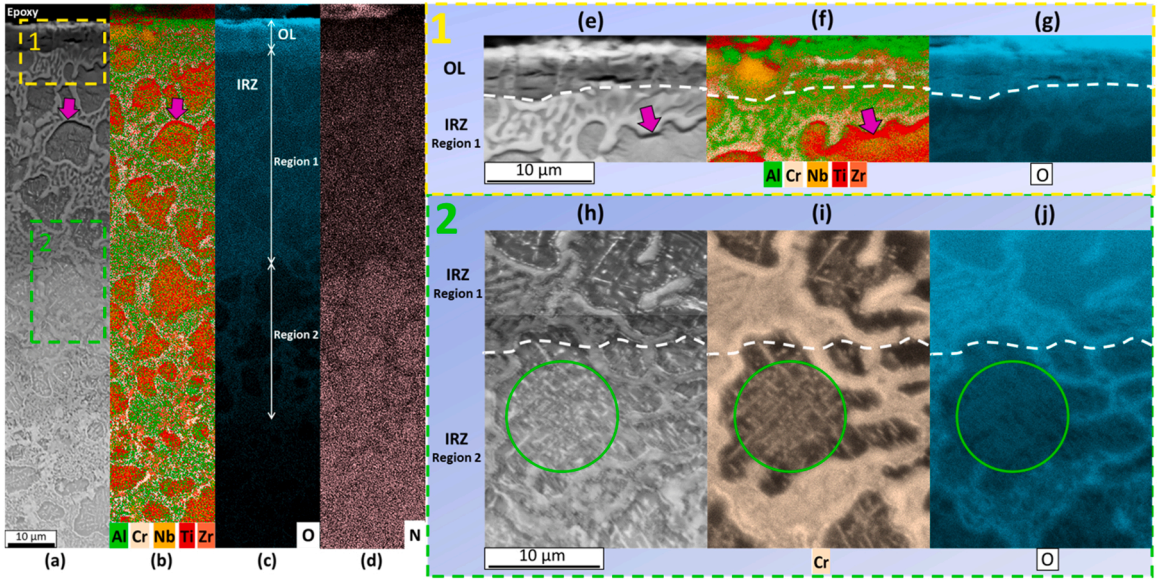


Fig. 3. Characterisation of a typical cross-section area of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy, oxidised at 800 °C, 100 h: (a, e, h) – SEM-BSE images ((a) – overall view of cross-section area; (e, h) – magnified images taken from zones 1 (e) and 2 (h) denoted in Fig. 3(a) with dashed yellow and green boxes, respectively); (b, f) – overlapped SEM-EDS maps of the constitutive elements (Al – green; Cr – beige; Nb – yellow; Ti – red; Zr – orange); (c, g, j) – SEM-EDS map of O distribution along the cross-section with denoted the oxide layer (OL) and the internal reaction zone (IRZ) composed of areas with higher (Region 1) and lower (Region 2) O content; (d) - SEM-EDS map of N distribution along the cross-section. Pink arrows in Fig. 3(a, b, e, f) illustrate typical Ti-rich segregations, located along the B2/Laves phase interfaces. Green circles in Fig. 3(h-j) highlight the preferential O-occupation of Cr-rich precipitates found in the internally reacted B2 phase. (For interpretation of the references to colour in this figure, the reader is referred to the web version of this article.)

Table 4
Average chemical compositions of the oxidation products in the OL of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy oxidised at 800 °C, 100 h. Similar products were also found in the OLs at 900 and 1000 °C. Effective (without counting oxygen) chemical compositions of the oxidation products are also given to highlight their preferential occupation by certain constitutive elements.

Oxidation product in the OL	Elements, at.%					
	O	Al	Cr	Nb	Ti	Zr
Al-rich	73.0	12.2	4.3	1.2	7.3	1.9
	–	44.9	16.0	4.4	27.4	7.3
Cr-rich	72.6	4.3	16.0	1.1	4.1	1.8
	–	16.2	57.9	4.2	14.8	6.9
(Al, Nb, Ti)-rich (completely reacted B2 phase)	73.5	6.2	3.0	6.1	9.7	1.5
	–	22.9	11.4	23.4	36.5	5.9
Ti-rich	71.9	4.0	4.7	3.0	15.4	1.1
	–	13.5	16.7	10.1	56.6	3.1
(Al, Zr)-rich (completely reacted C14 Laves phase)	70.7	9.9	4.7	2.3	6.3	6.2
	–	32.2	16.1	8.7	21.6	21.4

oxidation rate constant (mg/cm^2 h); t – time (h); n – the time exponent. Fitting of the 800–900 °C-curves by Eq. (1) resulted in $n = 0.58$–0.49 (Table 3). These values suggested that the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy obeyed near-parabolic oxidation kinetics at 800–900 °C. Rough approximation of the 1000 °C-curve gave $n = 0.52$, meaning the oxidation of the alloy also followed a near-parabolic law before spallation.

3.3. Structure after oxidation

3.3.1. Cross-section analysis after oxidation at 800 °C

To unveil the oxidation mechanisms of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy with the best spallation resistance, we first analysed cross-sections of its oxidised specimens. At 800 °C, a ~6 μm-depth adherent oxide layer (OL) (~ 73 at% of O) was formed (Fig. 3(a, c, e, g)); yet, tiny cracks along the surface should be noted (Fig. 3(e)). The overall constitution of the OL mimicked the starting hypoeutectic microstructure (Fig. 3(e, f)); however, some constitutive elements redistributed markedly, producing specific oxidation products. Particularly, thin, discontinuous Ti- and Al-rich layers covered the top of completely reacted (Al, Nb, Ti)-rich B2 and (Al, Zr)-rich Laves phase-based areas, respectively (Fig. 3(e-g); Table 4). Besides, tiny and rare Cr-rich islands were observed amid the Laves phase lamellae (Fig. 3(f); Table 4).

Beneath the OL, we found thin (Ti, N)-rich islands and a ~80 μm-depth internal reaction zone (IRZ) composed of two regions, morphologically similar to the initial microstructure (Fig. 3(a, e, d)). In Region 1 (~ 58 at% of O; ~ 45 μm), Ti segregated to the upper part of the internally reacted B2 regions along the B2/Laves phase interfaces, oriented parallel to the surface (highlighted with pink arrows in Fig. 3(a, b, e, f)). Interestingly, O attacked the Laves phase particles, while N invaded the B2 phase; this was more evident in Region 2 (~ 39 at% of O; ~ 35 μm) (Fig. 3(b-d)). However, the magnified EDS maps showed that O still presented in the B2 phase in Region 2, albeit it was distributed heterogeneously (Fig. 3(j)). Oxygen partitioned to dispersed and intersected needle-shaped Cr-rich precipitates, whilst the precipitate-free volumes of the B2 phase remained O-leaned (one of the typical zones

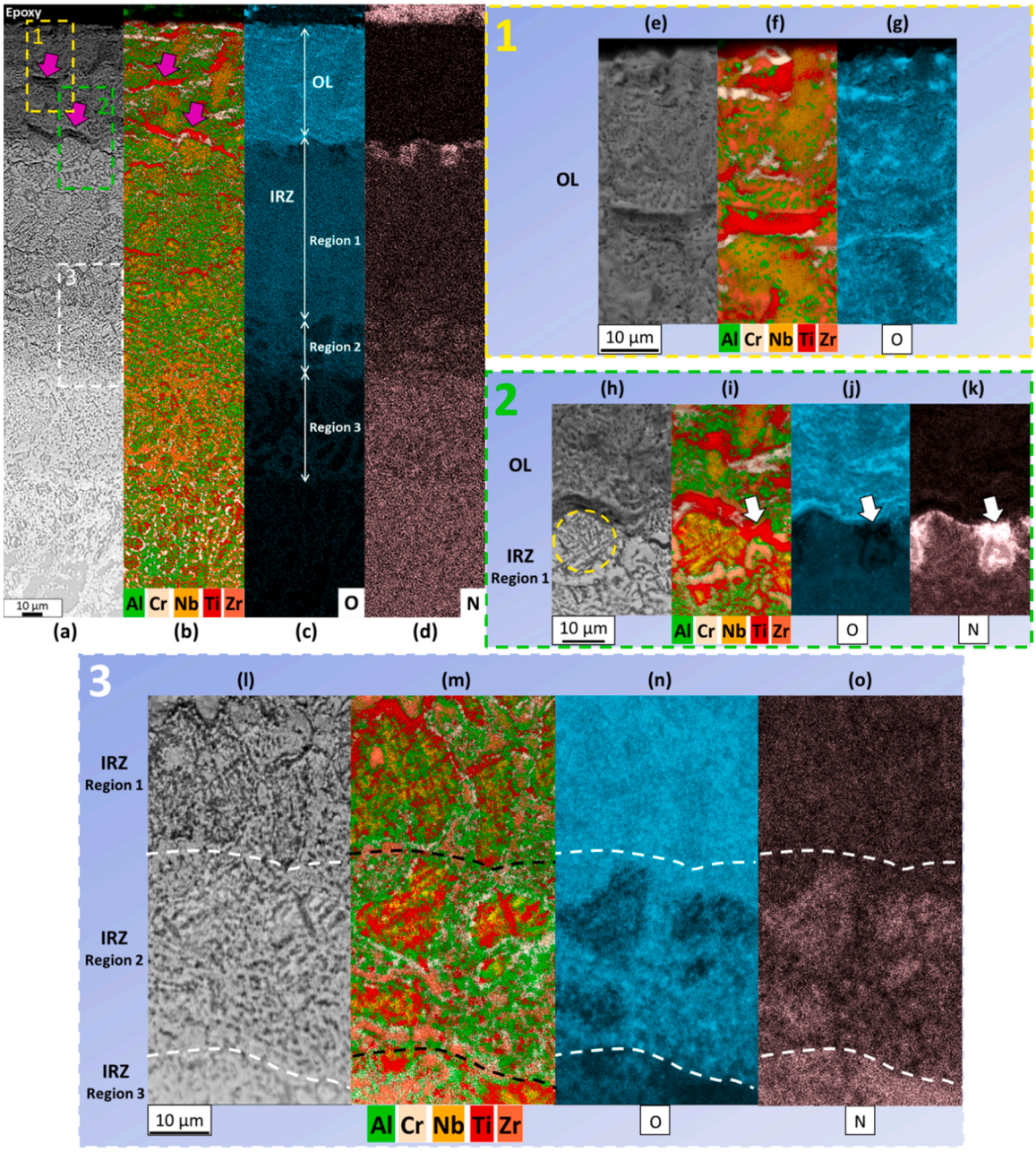


Fig. 4. Characterisation of a typical cross-section area of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy, oxidised at 900 °C, 100 h: (a, e, h, l) – SEM-BSE images ((a) – overall view of cross-section area; (e, h, l) – magnified images taken from zones 1 (e), 2 (h), and 3 (l) denoted in Fig. 4(a) with dashed yellow, green, and white boxes, respectively); (b, f, i, m) – overlapped SEM-EDS maps of the constitutive elements (Al – green; Cr – beige; Nb – yellow; Ti – red; Zr – orange); (c, g, j, n) – SEM-EDS map of O distribution along the cross-section with denoted the oxide layer (OL) and the internal reaction zone (IRZ) composed of regions with higher (Region 1) and lower (Regions 2 and 3) O content; (d, k, o) – SEM-EDS map of N distribution along the cross-section. Pink arrows in Fig. 4(a, b) illustrate typical Ti-rich segregations, located along the B2/Laves phase interfaces. Dashed yellow circle in Fig. 4(h) shows typical products, found in the internally reacted B2 phase. White arrows in Fig. 4 (i-k) display typical (Ti, N)-rich (and O-lean) pegs on the boundary between the OL and IRZ. (For interpretation of the references to colour in this figure, the reader is referred to the web version of this article.)

is highlighted with green circles in Fig. 3(h-j)). The Cr-rich particles were also found in the base material (Fig. 3(a, b)), which was barely affected by oxidation/nitridation. For instance, the bulk concentrations of O/N, determined by inert gas fusion method [41–43], in the as-cast and tested (an unoxidised core was used for the analysis) specimens were 0.11/0.013 at% and 0.17/0.016 at%, respectively. These data indicate that the appearance of the Cr-rich particles could stem from the thermal instability of the alloy rather than its oxidation/nitridation.

3.3.2. *Cross-section analysis after oxidation at 900 °C*

At 900 °C, a ~55 μm-depth adherent OL (~ 71 at% of O) was formed (Fig. 4(a)). Similar to the 800 °C-oxidation, Ti-rich layer covered the top of the OL, albeit assemblies of other elements were also observed. Meantime, the oxidation products in the inner part of the OL evolved compared to the 800 °C-oxidation (Fig. 4(a-k)). Specifically, Cr-rich islands/discrete layers became larger and they arranged parallel to the surface, close to alternating (Al, Zr)-rich clusters/lamellae. Interestingly, free spaces between the latter were filled with tiny but clearly discernable Al-rich particles (Fig. 4(e-j)). All the mentioned structural

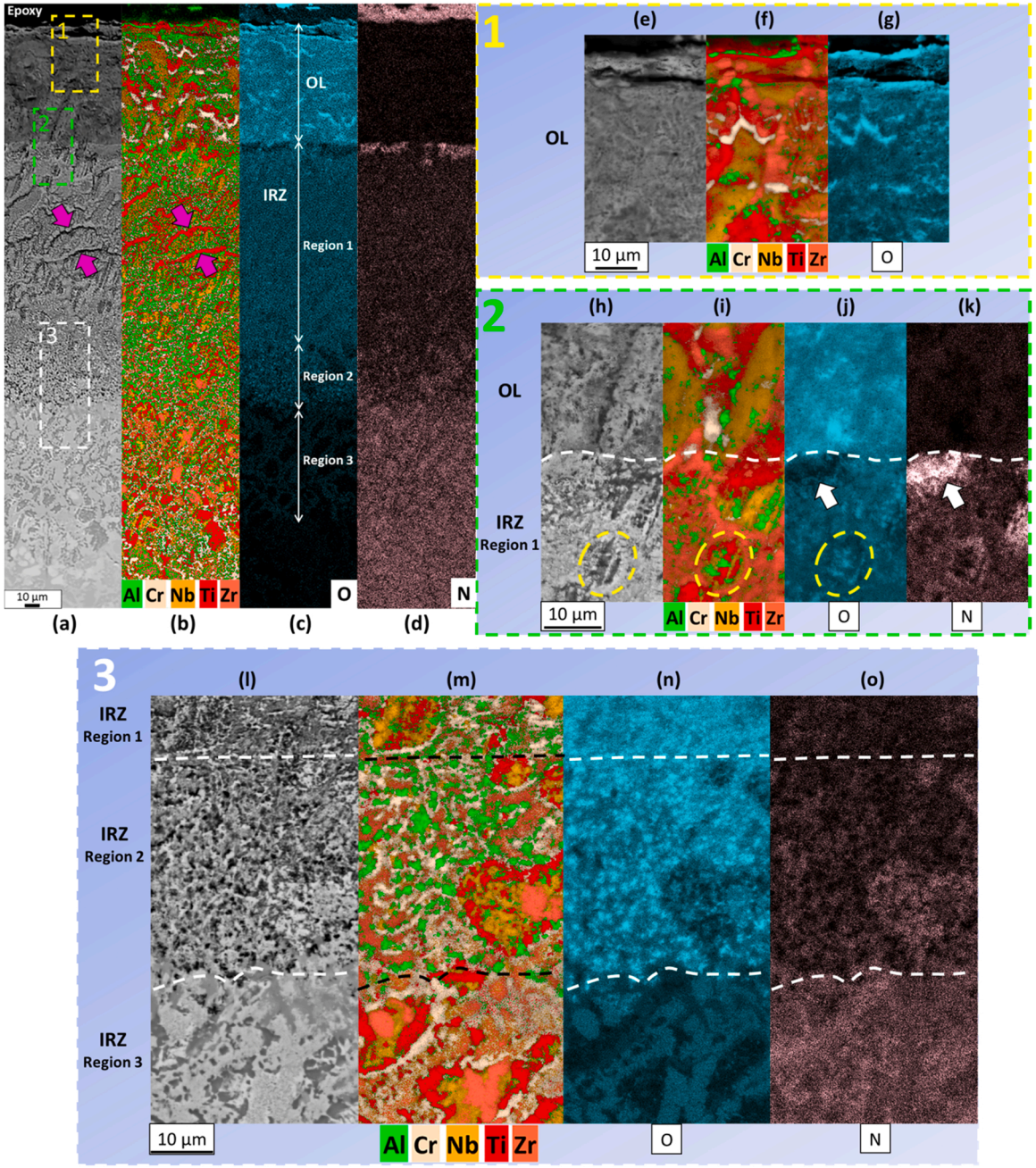


Fig. 5. Characterisation of a typical cross-section area of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy oxidised at 1000 °C, 10 h: (a, e, i, m) – SEM-BSE images ((a) – overall view of cross-section area; (e, i, m) – magnified images taken from zones 1 (e), 2 (i), and 3 (m) denoted in Fig. 5(a) with dashed yellow, green, and white boxes, respectively); (b, f, j, n) – overlapped SEM-EDS maps of the constitutive elements (Al – green; Cr – beige; Nb – yellow; Ti – red; Zr – orange); (c, g, k, o) – SEM-EDS map of O distribution along the cross-section with denoted the oxide layer (OL) and the internal reaction zone (IRZ) composed of regions with higher (Region 1) and lower (Regions 2 and 3) O content; (d, h, l, p) – SEM-EDS map of N distribution along the cross-section. Pink arrows in Fig. 5(a, c, d, e, h, i, l) illustrate typical Ti-rich segregations, located along the B2/Laves phases interfaces. Dashed yellow circles in Fig. 5(i-j) show typical products found in the internally reacted B2 phase. (For interpretation of the references to colour in this figure, the reader is referred to the web version of this article.)

constituents referred to completely reacted Laves phase-based eutectic areas. The bottom sides of these areas were decorated by relatively large Ti-rich clusters (denoted with pink arrows in Fig. 4(a, b)), segregated in the upper part of the former (Al, Nb, Ti)-rich B2 phase (similar to the 800 °C-oxidation), also containing dispersed Al-rich particles (Fig. 4(e-j)). The chemical compositions of the oxidation products in the OL were analogous to those of the 800 °C-oxidation (Table 4).

Under the OL, a ~210 μm-depth IRZ with several regions was observed (Fig. 4(a, c)). On the boundary between the OL and IRZ (Region 1 (~ 58 at% of O; ~90 μm)), (Ti, N)-rich (and O-lean) pegs were formed along the B2/Laves phase interfaces (highlighted with white arrows in Fig. 4(i-k)). Here, the internally reacted B2 phase, comprising (Al, Nb, Ti)-rich zones and lenticular/needle-shaped and intersected Ti-rich conglomerates, decorated with sparse and small Al-rich particles, was revealed (marked with the yellow circle in Fig. 4(h)). These products could be tracked throughout Region 1, having the homogeneous O-enrichment. In Region 2 (~ 49 at% of O; ~25 μm), N and O attacked the places, morphologically similar to the B2 and Laves phases, respectively. Yet, in the former B2 phase, similarly to the 800 °C-oxidation, heterogeneous distribution of O should be noted (Fig. 4(a-d, l-o)). In Region 2,

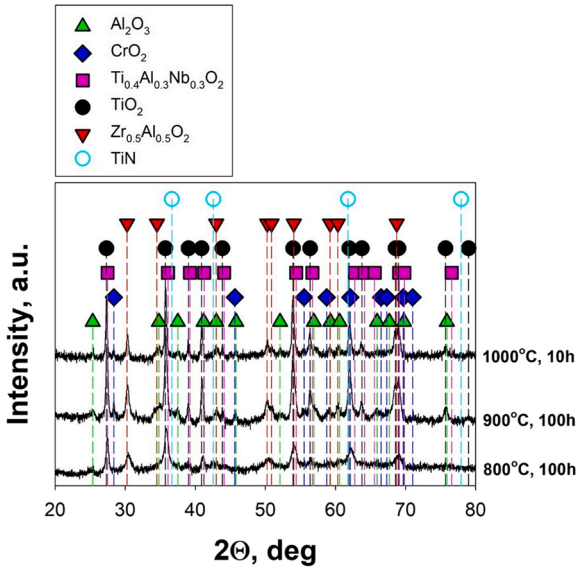


Fig. 6. XRD patterns collected from the surfaces of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy oxidised at 800–1000 °C.

the lenticular Ti-rich conglomerates were larger (Fig. 4(m)). The partitioning of O and N became even more visible in Region 3 (O content of ~ 31 at%; ~55 μm). Nitrogen still permeated into the alloy after the end of the oxidation front (~40 μm), producing multiple, relatively large round-shaped (Ti, N)-rich regions (Fig. 4(a-d)). They contained fine Al-rich clusters and were surrounded by thin Cr-rich layers (Fig. 4(a-d)). The base material underwent complex phase transformations (Fig. 4(a, b)).

3.3.3. Cross-section analysis after oxidation at 1000 °C

After oxidation at 1000 °C for 10 h, a ~75 μm-depth adherent, yet cracks/pores-containing, OL was formed (~ 73 at% of O) (Fig. 5(a)). On its top part, which tended to spall off, thin, almost continuous and repeated Al- and Ti-rich layers dominated (Fig. 5(a-c, e-g)); similar layers covered the edges of the crack(s) (Fig. 5(f, g)). Meanwhile, the inner part of the OL and a ~275 μm-depth IRZ resembled those of 900 °C (Fig. 5(a-o)). Specifically, we also observed the Ti-rich segregations along the B2/Laves phase interfaces, oriented parallel to the surface (denoted with pink arrows in Fig. 5(a, b)), and (Ti, N)-rich pegs on the boundary between the OL and IRZ (Fig. 5(k)). Again, the chemical compositions of the oxidation products in the OL were close to the 800 °C-oxidation (Table 4).

However, some discrepancies between the 900 °C and 1000 °C-oxidation should be noted. In the latter case, the lenticular Ti-rich conglomerates, alternating with (Al, Nb, Ti)-rich zones and decorated by Al-rich particles (a typical zone in Region 1 (~ 60 at% of O; ~130 μm) is highlighted with dashed yellow circles in Fig. 5(h-k)), within the internally reacted B2 phase were predominantly parallel and thicker in contrast to those of 900 °C. In Region 2 (~ 51 at% of O; ~40 μm), these conglomerates appeared to be round-shaped (Fig. 5(m)).

Similar to the 900 °C-oxidation, Region 2 was the last zone where the large volume fraction of the Al-rich particles could be found (Fig. 5(b, m)). Beneath, Zr-rich phase(s) dominated (Region 3 (~ 31 at% of O; ~75 μm)) till the end of the oxidation front (Fig. 5(b, m)). Nitridation was slightly deeper (~30 μm), leading to the exclusive formation of relatively large (Ti, N)-rich areas (Fig. 5(b, d)). In the base material, massive phase transformations should also be noted (Fig. 5(a, b)).

3.3.4. Surface analysis

To identify the oxidation products formed in the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy at 800–1000 °C, we conducted XRD analysis of the oxidised surfaces (Fig. 6). Examination of XRD patterns revealed five different oxides: four tetragonal-type and one trigonal-type (Fig. 6; Table 5). Surveying of existing oxide prototypes with close chemical compositions (Table 4) and lattice parameters allowed us to identify the Al-rich, Cr-rich, (Al, Nb, Ti)-rich, Ti-rich, and (Al, Zr)-rich products as Al_2O_3, CrO_2, $Ti_{0.4}Al_{0.3}Nb_{0.3}O_2$, TiO_2, and $Zr_{0.5}Al_{0.5}O_2$ oxides, respectively. Differencies in the lattice parameters of the prototype and experimental oxides could be ascribed to more complex chemical compositions of the latter. Specifically, the designation of Al-rich, Cr-rich, and Ti-rich oxides as Al_2O_3, CrO_2, and TiO_2, respectively, was only made to accentuate their structure types and did not reflect the actual chemical compositions. Note, Bragg's peaks corresponded to the TiO_2 had the highest intensity, indicating its dominance over the other oxides. Also, several low-intensity Bragg's peaks belonged to TiN (Space group 225; a = 0.4241 nm; [44]) were found. Interestingly, similar products were observed in the oxidised remnants of the $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$ and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys (Fig. S1, Supplementary Material).

4. Discussion

4.1. Spallation resistance

One of the notable results of our work is the different spallation resistance of the studied alloys. The hypoeutectic $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy sustained cyclic oxidation (we assume that the interrupted testing scheme chosen in this work could be considered as cyclic oxidation with proportionally increased exposures between the cycles) at 800–1000 °C much better than the eutectic $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$ and the hypereutectic $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys (Fig. 2). In contrast to the spallation of only oxide scales found in other Al-Cr-Nb-Ti-Zr RHEAs [31,32] or pesting phenomena in a $Hf_{0.5}Nb_{0.5}Ta_{0.5}Ti_{1.5}Zr$ RHEA [50], the $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$ and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys fragmented after a certain period at every cooling cycle, exposing the unoxidised core, and this process accelerated with the temperature increment, that is, these alloys showed thermal shock response typical for brittle materials [51]. Considering the similarity of oxidation products (i.e. the equality of growth stresses in the oxide scales), different spallation resistance of the studied alloys could be explained by the structural factors. Due to the abundance of the more brittle (than the B2 phase) C14 Laves phase, the $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$ and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys relieved high thermal stresses, accumulated during cooling and originated from thermal shock rather than mismatched thermal expansion coefficients of

Table 5
Structure types, space groups, and lattice parameters of the prototype and experimental oxides.

Structure type	Space group	Prototype oxide	Lattice parameters, nm			
			prototype		experimental	
			a	c	a	c
Trigonal	167	Al_2O_3 [45]	0.4760	1.300	0.4792	1.314
Tetragonal	136	CrO_2 [46]	0.4420	0.2930	0.4442	0.2986
Tetragonal	136	$Ti_{0.4}Al_{0.3}Nb_{0.3}O_2$ [47]	0.4587	0.2960	0.4589	0.2958
Tetragonal	136	TiO_2 [48]	0.4600	0.2963	0.4612	0.2993
Tetragonal	137	$Zr_{0.5}Al_{0.5}O_2$ [49]	0.3591	0.5189	0.3587	0.5192

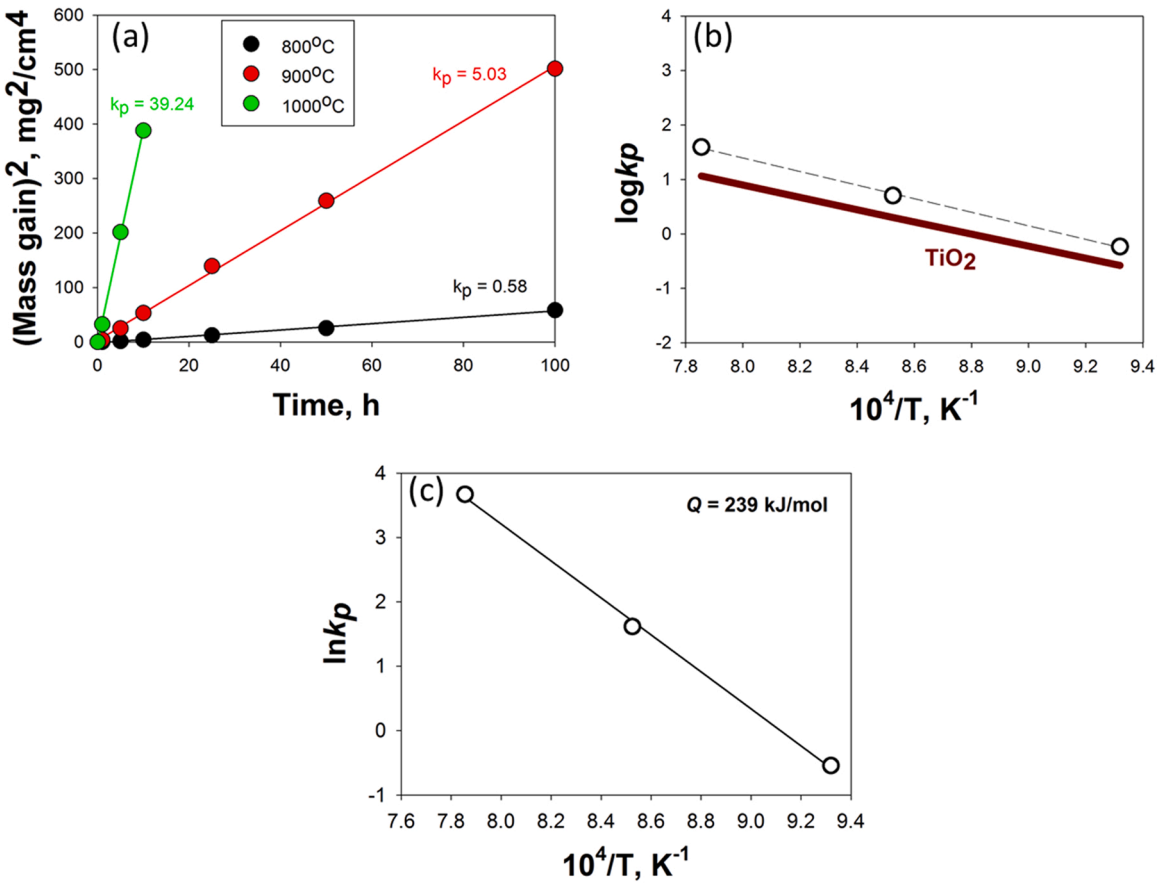


Fig. 7. Characterisation of oxidation kinetics of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy: (a) – a plot $(\Delta m)^2 - t$, showing the k_p values for 800, 900, and 1000 °C; (b) – a logkp – 10^4/T plot, illustrating the similarity of the obtained k_p and the k_p of TiO_2; (c) a lnk_p – 10^4/T plot, used for evaluation of the Q.

the oxide scale and bulk material [51], by the interface cracking. The latter process provoked first the oxide scale spallation and then the fragmentation/fracture of the bulk material, as the surface cracks easily propagated inside. The better spallation/thermal shock resistance of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy could stem from the higher volume fraction of the softer (than the C14 Laves phase) B2 phase (Table 2). This phase, most likely, blocked the nucleation or hindered the growth and coalescence of the thermally induced cracks during the testing at 800–900 °C, but had limited efficiency at 1000 °C.

4.2. Oxidation kinetics and oxidation mechanism

For the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy, showing near-parabolic oxidation at 800–1000 °C, we estimated the parabolic oxidation rate constants, k_p (Fig. 7(a)). The obtained values were almost similar to k_p of the pure TiO_2 [52] (Fig. 7(b)). To analyse the rate-controlling mechanism of the oxide layer thickening, the Arrhenius equation was applied [53]:

$$k_p = Aexp(-Q/RT) \quad (2)$$

where k_p – the parabolic rate oxidation constant (Fig. 7(a)), A – a constant, R – gas constant, Q – activation energy for the oxidation. A linear approximation of the plot lnk_p – 10^4/T resulted in Q = 239 kJ/mol (Fig. 7(c)). This value correlated well with the diffusion activation energy of O in Ti (Q = 246 kJ/mol [54]), and O (Q = 234 kJ/mol [55]) or Ti (Q = 257 kJ/mol [56]) in the TiO_2. Thus, the oxide scale growth in the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy could be simultaneously governed by the inward diffusion of O and outward diffusion of Ti.

Indeed, Ti and O diffusion paths in the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy can be tracked in the SEM-EDS maps (Figs. 4–6). Ti tended to move outwardly from the B2 phase and form the TiO_2, which dominated on the top of the OL at all the temperatures. In the inner part of the OL or in the IRZ, Ti agglomerated along the B2/Laves phase interfaces, perpendicular to its motion. The Laves phase arranged in such a fashion could retard the extraction of Ti, slowing the growth of the OL. However, this phase did not block the O penetration. Conversely, it served as channels for a deeper O ingress than the B2 phase, aggravating the internal oxidation. The analogous interaction of the Laves phase with O was recently reported by Huang et al. [57].

In turn, not only diffusion of Ti and O was observed; Cr also diffused. Cr preferentially occupied the Laves phase in the initial state, but, when reacting with O, it formed the CrO_2 oxides. Here, we should remind that the application of simple formula, CrO_2, for highlighting of the Cr-rich oxide emphasised its rutile-type structure rather than the real chemical composition. In a pure form, the CrO_2 oxide is usually unstable at high temperatures. Meanwhile, traces of this oxide were previously identified by Seal et al. [58] in IN-738LC superalloy and Waseem et al. [32] in $Al_{12.5}Cr_{25}Mo_xNb_{25}Ti_{25}Zr_{12.5}$ alloys oxidised at 900 °C and 1000 °C, respectively. The nucleation of non-stable oxides was also reported by Butler et al. in CrNbTiZr [26] and CrNbTiTa [59] alloys. They assumed that NbO oxides could form internally where the oxygen partial pressure was considerably reduced compared to the free surface [59]. In the studied alloys, the stability of the CrO_2 was suggested to originate from its complex chemical composition (Table 4; [60]) and the presence of a mixture of multiple oxides (Figs. 3–6; Table 4; [58]). However, the preferential formation of the rutile-type CrO_2 over a more expected trigonal-type Cr_2O_3 oxide requires an additional in-depth study.

In the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy, the CrO_2 nucleated predominantly along the B2/Laves phase interfaces; note that, at 800 °C, the CrO_2 oxides were also found in the core of the B2 phase. We assume that such specific locations of the CrO_2 both in the OL and IRZ could stem from the

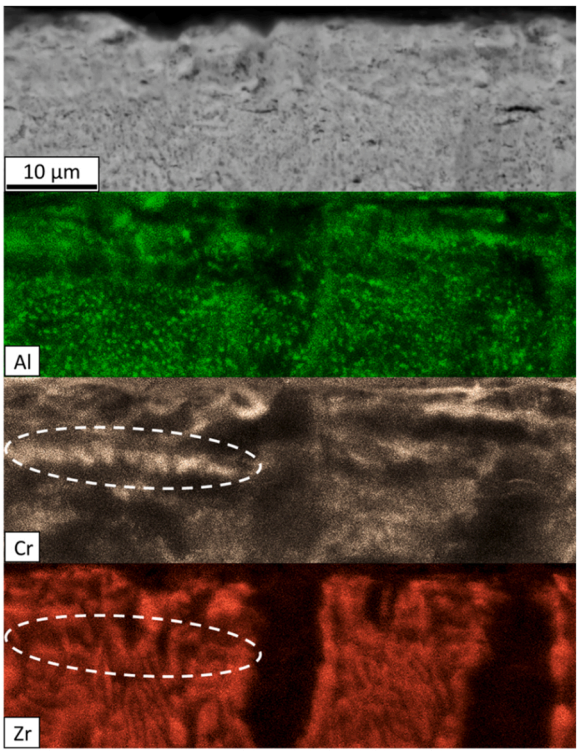


Fig. 8. Magnified image of the OL in the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy after oxidation at 900 °C, 100 h: a combined SEM-EDS image, showing the distribution of Al, Cr, and Zr. White dashed circles in Cr and Zr maps illustrate the discontinuous CrO_2 layer formed in the eutectic areas with the vertical orientation of the B2/Laves phase lamellae.

thermal instability of the alloy, leading to the prior precipitation of profuse Cr-rich nanoparticles then further oxidised. In contrast to the IRZ, the CrO_2 showed the ability to coagulate and grow in the OL laterally (Fig. 8). For the CrO_2 formed in the eutectic areas, this process depended on the B2/Laves phase lamellae orientation. Specifically, when being arranged perpendicularly to the surface, the Laves phase lamellae blocked the coalescence of the CrO_2, located in the neighbour interphases, and, thus, prevented the creation of a continuous layer (typical regions denoted with white dashed ellipses in Fig. 8). This discontinuous morphology made the CrO_2 ineffective against oxidation [32].

Another interesting feature was the formation of the Al_2O_3 oxides. Similar to the CrO_2, the Al_2O_3 nucleated in the primary and lamellar B2 phases, but the distribution was more homogeneous (Fig. 8). Nevertheless, once formed, the dispersed Al_2O_3 oxides could not agglomerate at 800–900 °C to provide adequate protection against O attack (Figs. 4, 5, and 8). Signs of the continuous Al_2O_3 layer started to appear on the top of the OL only at 1000 °C (Fig. 6). Besides, two distinct zones with the high volume fraction of the discrete Al_2O_3 were also formed at 900–1000 °C just beneath the OL and at the bottom part of the IRZ when the oxidation front weakened. In these zones, the nucleation of the Al_2O_3 seemed to be more selective than in the OL. Particularly, the Al_2O_3 preferentially decorated the lenticular TiO_2 within the B2 phase. Judging from the specific shape of these internally reacted products, they could originate from the prior precipitation of a Ti_3Al phase [21], which subsequently oxidised to the TiO_2 and the Al_2O_3.

In addition to a variety of complex oxides found in the Al-Cr-Nb-Ti-(V)-Zr RHEAs [26,27,30–34], we detected some new species, namely the $Ti_{0.4}Al_{0.3}Nb_{0.3}O_2$ and the $Zr_{0.5}Al_{0.5}O_2$ (Table 4; Fig. 6; Table 5). The $Ti_{0.4}Al_{0.3}Nb_{0.3}O_2$ was a rutile-type oxide, formed beneath the TiO_2 in the bottom part of the completely reacted B2 phase (Fig. 3(f), 4(f), 5(f)). Meanwhile, the $Zr_{0.5}Al_{0.5}O_2$ was an Al-doped version of a ZrO_2, nucleated during the O attack of the C14 Laves phase (Fig. 3(f), 4(f), 5(f), 8). Relying on recent studies [8,9,26,61,62], claiming the complex rutile-type oxides to ensure fair protection against oxidation, we speculate that the $Ti_{0.4}Al_{0.3}Nb_{0.3}O_2$ acted similarly. However, the exact influence of the $Ti_{0.4}Al_{0.3}Nb_{0.3}O_2$ and the $Zr_{0.5}Al_{0.5}O_2$ on the oxidation resistance needs to be investigated further.

Lastly, our study showed that, besides the oxidation, the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy experienced selective nitridation. Being undetected in the OL, nitrogen invaded exclusively the B2 phase in the IRZ. The preferential location of N in a bcc phase was also reported by Butler et al. [59]. They [59] mentioned that nitrides could be first to nucleate and further transformed to oxides with prolonged exposure. In our case, multiple discrete TiN pegs were formed in the upper part of the IRZ (just beneath the OL). Meanwhile, in the bottom part of the IRZ, the N permeation, which was slightly deeper than the O attack, created relatively large TiN regions. Nevertheless, the penetration depth of N (after the end of the oxidation front) was relatively small that can be deduced from the similarity of its bulk contents before and after oxidation tests (see Section 3.3.1.).

4.3. Comparison with other Al-Cr-Nb-Ti-(V)-Zr RHEAs and suggestions for future studies

Fig. 9 displays the mass gains of the investigated alloys and some other Al-Cr-Nb-Ti-(V)-Zr RHEAs. The temperature-time conditions for the analysis were chosen to cover most of the available literature data [26,27,30–34]. At 800 °C (Fig. 9(a)), all the three studied alloys showed the lowest mass gains amongst the selected counterparts. Particularly, the most oxidation-resistant $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloy gained only ~0.7 mg/cm^2, which was 11 times lower than the equiatomic $Al_{20}Cr_{20}Nb_{20}Ti_{20}Zr_{20}$ alloy [31]. After 1 h at 1000 °C (Fig. 9(b)), two ($Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$) of the studied alloys still outperformed a large part of the Al-Cr-Nb-Ti-(V)-Zr RHEAs, but they were inferior to the $Al_{12.5}Cr_{25}Nb_{25}Ti_{25}Zr_{12.5}$ alloy [32]. Only the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy could compete at a longer (10 h) time at 1000 °C with a markedly reduced number of the Al-Cr-Nb-Ti-(V)-Zr RHEAs (Fig. 9(c)). The $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy demonstrated better oxidation resistance than the $Al_{25}Nb_{25}Ti_{25}Zr_{25}$ [30] and $Cr_{25}Nb_{25}Ti_{25}Zr_{25}$ [26] alloys, but it was surpassed by the $Al_{20}Cr_{20}Nb_{20}Ti_{20}Zr_{20}$ and $Al_{12.5}Cr_{25}Nb_{25}Ti_{25}Zr_{12.5}$ alloys.

From these data, we can deduce a direct correlation between the oxidation resistance of the studied $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$, $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$, and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys and their chemical compositions: the higher the Al (and the lower Ti) content, the lower the mass gain. However, the increase in Al/Ti had a severe side effect: it deteriorated spallation resistance (Fig. 2; Section 4.1).

Meanwhile, establishing analogous relationships in the entire compositional space of Al-Cr-Nb-Ti-(V)-Zr RHEAs is a non-trivial task. Among the constitutive elements of the selected alloys, Al, Cr, and Zr were the most variable, while Nb and Ti remained almost constant. Specifically, the three most oxidation-resistant alloys at 1000 °C (20 h), namely $Al_{30}Cr_{10}Nb_{20}Ti_{20}Zr_{20}$, $Al_{20}Cr_{20}Nb_{20}Ti_{20}Zr_{20}$, and $Al_{12.5}Cr_{25}Nb_{25}Ti_{25}Zr_{12.5}$, were obtained mainly by Al, Cr, and Zr alterations (Table 1; Fig. 9(c)). Interestingly, reduced Al and Zr content in the $Al_{12.5}Cr_{25}Nb_{25}Ti_{25}Zr_{12.5}$ alloy resulted in lower mass gain at longer exposures (50 h) at 1000 °C. Moreover, recent work by Butler et al. [59] hinted at much better oxidation resistance of Al- and Zr-free CrNbTi alloy at 1200 °C compared to equiatomic $Al_{20}Cr_{20}Nb_{20}Ti_{20}Zr_{20}$ RHEA [31]. This can be explained partly by the absence of non-protective Zr-rich oxides [31]. In turn, Zeng et al. [63] showed very lately that high Al additions diminished the oxidation resistance of the CrAl coatings at 1200 °C. It was attributed to accelerated diffusion of Al, leading to the formation of discontinuous and unprotective Al_2O_3/Cr_2O_3 layers that tended to spall off.

These data show that there is still much work to be done before finding the Al-Cr-Nb-Ti-(V)-Zr RHEA with good oxidation and spallation

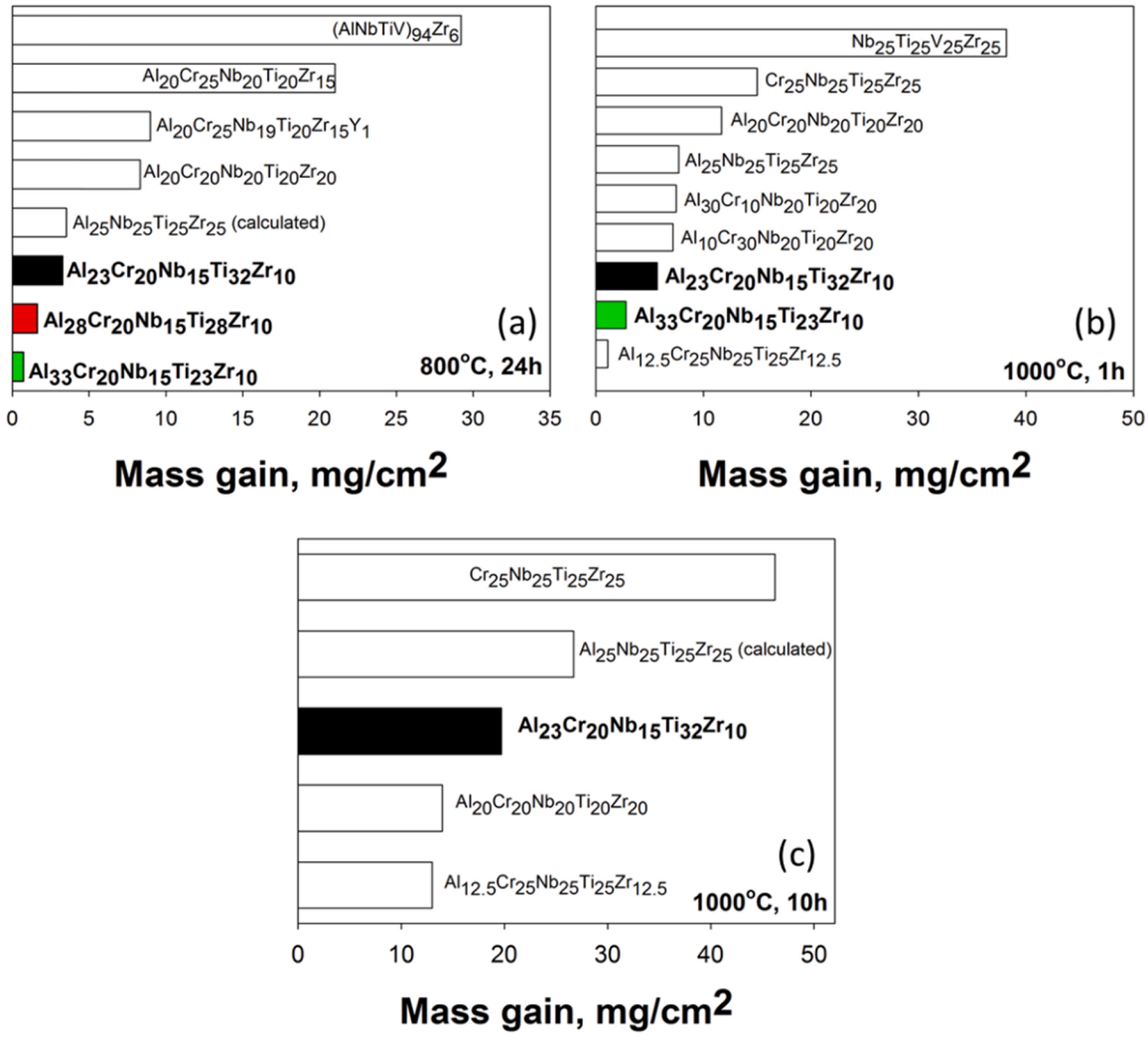


Fig. 9. Mass gain comparisons of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$, $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$, and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys with the Al-Cr-Nb-Ti-(V)-Zr RHEAs [26,27,30–34]: (a) – mass gains after 24 h at 800 °C; (b) – mass gains after 1 h at 1000 °C; (c) – mass gains after 10 h at 1000 °C.

resistance. Firstly, studies that will inspect the oxidation behaviour of the alloys during interrupted/cyclic tests are highly demanded because such conditions are close to practical usage. Secondly, the influence of the Al/Cr ratio is needed to be analysed more systematically. We assume that it can be tailored to satisfy both lower mass gain and improved oxide scale adherence [63]. Lastly, microalloying with rare earth elements should be further investigated as previous works [33,64] claimed the positive effect.

5. Conclusions

In this work, we studied the oxidation behaviour of the B2/Laves phase hypoeutectic $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$, eutectic $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$, and hypereutectic $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ RHEAs during the interrupted tests in the air at 800–1000 °C and exposures up to 100 h. The following conclusions were made:

1) The $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy demonstrated higher mass gains than the $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$ and the $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys, while it had much better spallation resistance due to the larger volume fraction of the B2 phase, which was softer than the C14 Laves phase. Compared to the current Al-Cr-Nb-Ti-(V)-Zr RHEAs, the studied alloys showed excellent oxidation resistance at 800 °C and remained competitive at 1000 °C.
2) The $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy obeyed near-parabolic ($n = 0.58$–0.52) kinetics at 800–1000 °C. The parabolic rate oxidation constants at 800, 900, and 1000 °C were 0.58, 5.03, and 39.24 mg^2/cm^4 h, respectively. These values were close to those of a pure TiO_2.
3) The TiO_2 dominated in the top of the oxide scales of the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy at 800–1000 °C. Additional Al_2O_3, CrO_2, $Ti_{0.4}Al_{0.3}Nb_{0.3}O_2$, and $Zr_{0.5}Al_{0.5}O_2$ were formed. Similar oxidation products were found in the $Al_{28}Cr_{20}Nb_{15}Ti_{27}Zr_{10}$ and $Al_{33}Cr_{20}Nb_{15}Ti_{22}Zr_{10}$ alloys.
4) Microstructure observations and oxidation activation energy calculations revealed that the outward diffusion of Ti and inward diffusion of O controlled the growth of the oxide layer.
5) Different behaviour of constitutive phases during oxidation was disclosed in the $Al_{23}Cr_{20}Nb_{15}Ti_{32}Zr_{10}$ alloy. The formation of the Al_2O_3, CrO_2, $Ti_{0.4}Al_{0.3}Nb_{0.3}O_2$, and TiO_2 was preferred in the B2 phase, while the Laves phase stood as a reservoir for the $Zr_{0.5}Al_{0.5}O_2$. Besides, the Laves phase impeded the outward diffusion of Ti, albiet aggravating the O ingress. The B2 phase, almost unaffected by O in the internally reacted zones, underwent the N invasion, creating TiN.

CRediT authorship contribution statement

N. Yurchenko: Conceptualization, Methodology, Investigation,

Validation, Visualisation, Formal analysis, Supervision, Writing – original draft, Writing – review & editing. **E. Panina**: Investigation, Validation. **S. Zherebtsov**: Writing – review & editing; **N. Stepanov**: Supervision, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgements

The authors gratefully acknowledge the financial support from the Russian Science Foundation Grant no. 19-79-30066 (https://rscf.ru/en/project/19-79-30066/). The work was carried out using the equipment of the Joint Research Center of Belgorod State National Research University "Technology and Materials" with financial support from the Ministry of Science and Higher Education of the Russian Federation within the framework of agreement № 075-15-2021-690 (unique identifier for the project RF——2296.61321×0030).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.corsci.2022.110464.

References

[1] J.H. Perepezko, The hotter the engine, the better, Science 326 (2009) 1068–1069, https://doi.org/10.1126/science.1179327.
[2] O.N. Senkov, G.B. Wilks, D.B. Miracle, C.P. Chuang, P.K. Liaw, Refractory high-entropy alloys, Intermetallics 18 (2010) 1758–1765, https://doi.org/10.1016/j.intermet.2010.05.014.
[3] O.N. Senkov, D.B. Miracle, K.J. Chaput, J.P. Couzinie, Development and exploration of refractory high entropy alloys - A review, J. Mater. Res. 33 (2018) 3092–3128, https://doi.org/10.1557/jmr.2018.153.
[4] O.N. Senkov, G.B. Wilks, J.M. Scott, D.B. Miracle, Mechanical properties of $Nb_{25}Mo_{25}Ta_{25}W_{25}$ and $V_{20}Nb_{20}Mo_{20}Ta_{20}W_{20}$ refractory high entropy alloys, Intermetallics 19 (2011) 698–706, https://doi.org/10.1016/j.intermet.2011.01.004.
[5] M. Wang, Z. Ma, Z. Xu, X. Cheng, Microstructures and mechanical properties of HfNbTaTiZrW and HfNbTaTiZrMoW refractory high-entropy alloys, J. Alloy. Compd. 803 (2019) 778–785, https://doi.org/10.1016/j.jallcom.2019.06.138.
[6] O.N. Senkov, S. Gorsse, D.B. Miracle, High temperature strength of refractory complex concentrated alloys, Acta Mater. 175 (2019) 394–405, https://doi.org/10.1016/J.ACTAMAT.2019.06.032.
[7] Q. Wei, G. Luo, R. Tu, J. Zhang, Q. Shen, Y. Cui, Y. Gui, A. Chiba, High-temperature ultra-strength of dual-phase $Re_{0.5}MoNbW(TaC)_{0.5}$ high-entropy alloy matrix composite, J. Mater. Sci. Technol. (2021), https://doi.org/10.1016/j.jmst.2020.12.015.
[8] K.C. Lo, Y.J. Chang, H. Murakami, J.W. Yeh, A.C. Yeh, An oxidation resistant refractory high entropy alloy protected by $CrTaO_4$-based oxide, Sci. Rep. 9 (2019) 1–12, https://doi.org/10.1038/s41598-019-43819-x.
[9] B. Gorr, F. Müller, S. Schellert, H.-J. Christ, H. Chen, A. Kauffmann, M. Heilmaier, A new strategy to intrinsically protect refractory metal based alloys at ultra high temperatures, Corros. Sci. (2020), 108475, https://doi.org/10.1016/j.corsci.2020.108475.
[10] K.C. Lo, H. Murakami, J.W. Yeh, A.C. Yeh, Oxidation behaviour of a novel refractory high entropy alloy at elevated temperatures, Intermetallics 119 (2020), 106711, https://doi.org/10.1016/j.intermet.2020.106711.
[11] S. Wu, D. Qiao, H. Zhang, J. Miao, H. Zhao, J. wang, Y. Lu, T. Wang, T. Li, Microstructure and mechanical properties of $C_xHf_{0.25}NbTaW_{0.5}$ refractory high-entropy alloys at room and high temperatures, J. Mater. Sci. Technol. (2021), https://doi.org/10.1016/j.jmst.2021.05.015.
[12] F. Appel, M. Paul, D.H. Oehring, Gamma Titanium Aluminide Alloys, Wiley-VCH Verlag & Co. KGaA, 2011.
[13] O.N. Senkov, S.V. Senkova, D.B. Miracle, C. Woodward, Mechanical properties of low-density, refractory multi-principal element alloys of the Cr–Nb–Ti–V–Zr system, Mater. Sci. Eng. A. 565 (2013) 51–62, https://doi.org/10.1016/j.msea.2012.12.018.
[14] N.D. Stepanov, D.G. Shaysultanov, G.A. Salishchev, M.A. Tikhonovsky, Structure and mechanical properties of a light-weight AlNbTiV high entropy alloy, Mater. Lett. 142 (2015) 153–155, https://doi.org/10.1016/j.matlet.2014.11.162.
[15] N.D. Stepanov, N.Y. Yurchenko, D.V. Skibin, M.A. Tikhonovsky, G.A. Salishchev, Structure and mechanical properties of the $AlCr_xNbTiV$ (x = 0, 0.5, 1, 1.5) high entropy alloys, J. Alloy. Compd. 652 (2015) 266–280, https://doi.org/10.1016/j.jallcom.2015.08.224.
[16] N.Y. Yurchenko, N.D. Stepanov, D.G. Shaysultanov, M.A. Tikhonovsky, G. A. Salishchev, Effect of Al content on structure and mechanical properties of the $Al_xCrNbTiVZr$ (x = 0; 0.25; 0.5; 1) high-entropy alloys, Mater. Charact. 121 (2016) 125–134, https://doi.org/10.1016/j.matchar.2016.09.039.
[17] N.Y. Yurchenko, N.D. Stepanov, S.V. Zherebtsov, M.A. Tikhonovsky, G. A. Salishchev, Structure and mechanical properties of B2 ordered refractory $AlNbTiVZr_x$ (x = 0–1.5) high-entropy alloys, Mater. Sci. Eng. A. 704 (2017) 82–90, https://doi.org/10.1016/j.msea.2017.08.019.
[18] N.D. Stepanov, N.Y. Yurchenko, E.S. Panina, M.A. Tikhonovsky, S.V. Zherebtsov, Precipitation-strengthened refractory $Al_{0.5}CrNbTi_2V_{0.5}$ high entropy alloy, Mater. Lett. 188 (2017) 162–164, https://doi.org/10.1016/J.MATLET.2016.11.030.
[19] N.Y. Yurchenko, E.S. Panina, S.V. Zherebtsov, M.A. Tikhonovsky, G.A. Salishchev, N.D. Stepanov, Microstructure evolution of a novel low-density Ti–Cr–Nb–V refractory high entropy alloy during cold rolling and subsequent annealing, Mater. Charact. 158 (2019), https://doi.org/10.1016/j.matchar.2019.109980.
[20] N. Yurchenko, E. Panina, M. Tikhonovsky, G. Salishchev, S. Zherebtsov, N. Stepanov, Structure and mechanical properties of an in-situ refractory $Al_{20}Cr_{10}Nb_{15}Ti_{20}V_{25}Zr_{10}$ high entropy alloy composite, Mater. Lett. (2020), 127372, https://doi.org/10.1016/j.matlet.2020.127372.
[21] N. Yurchenko, E. Panina, S. Zherebtsov, N. Stepanov, Design and characterization of eutectic refractory high entropy alloys, Materialia 16 (2021), 101057, https://doi.org/10.1016/j.mtla.2021.101057.
[22] N.Y. Yurchenko, E.S. Panina, G.A Salishchev, N.D. Stepanov, Design and characterization of Al-Cr-Nb-Ti-V-Zr high-entropy alloys for high-temperature applications, Phys. Mesomech. 24 (2021) 642–652, https://doi.org/10.1134/S1029959921060023, 2021 246.
[23] X.W. Liu, Z.C. Bai, X.F. Ding, J.Q. Yao, L. Wang, Y.Q. Su, Z.T. Fan, J.J. Guo, A novel light-weight refractory high-entropy alloy with high specific strength and intrinsic deformability, Mater. Lett. 287 (2021), 129255, https://doi.org/10.1016/j.matlet.2020.129255.
[24] L. Li-Yan, C. Shu-nan, L. Yi, J. Gang, C. Hai-Dong, J. Yun-Jie, L. Chang-Jiu, L. Cheng-Xin, Microstructure and mechanical properties of lightweight $Al_xCrNbTiV$ (x = 0.2, 0.5, 0.8) refractory high entropy alloys, Int. J. Refract. Met. Hard Mater. (2022), 105784, https://doi.org/10.1016/J.IJRMHM.2022.105784.
[25] R. Huang, J. Tan, W. Li, Q. Dong, C.J. Li, X.M. Qin, S.F. Guo, Y.P. Lu, Effects of V content on the microstructure and mechanical properties of $Nb_{31}Ti_{37-x}Zr_{26}Al_6V_x$ refractory medium-entropy alloys, Intermetallics 143 (2022), 107472, https://doi.org/10.1016/J.INTERMET.2022.107472.
[26] T.M. Butler, K.J. Chaput, J.R. Dietrich, O.N. Senkov, High temperature oxidation behaviors of equimolar NbTiZrV and NbTiZrCr refractory complex concentrated alloys (RCCAs), J. Alloy. Compd. 729 (2017) 1004–1019, https://doi.org/10.1016/j.jallcom.2017.09.164.
[27] N. Yurchenko, E. Panina, S. Zherebtsov, G. Salishchev, N. Stepanov, N. Yurchenko, E. Panina, S. Zherebtsov, G. Salishchev, N. Stepanov, Oxidation Behavior of Refractory AlNbTiVZr0.25 High-Entropy Alloy, in: Materials, 11, 2018, p. 2526, https://doi.org/10.3390/ma11122526.
[28] M. Esmaily, Y. Qiu, S. Bigdeli, M.B. Venkataraman, A. Allanore, N. Birbilis, High-temperature oxidation behaviour of $Al_xFeCrCoNi$ and AlTiVCr compositionally complex alloys, Npj Mater. Degrad. 41 (4) (2020) 1–10, https://doi.org/10.1038/s41529-020-00129-2, 2020.
[29] X. Shi, W. Yang, Z. Cheng, W. Shao, D. Xu, Y. Zhang, J. Chen, Influence of micro arc oxidation on high temperature oxidation resistance of AlTiCrVZr refractory high entropy alloy, Int. J. Refract. Met. Hard Mater. 98 (2021), 105562, https://doi.org/10.1016/j.ijrmhm.2021.105562.
[30] J. Jayaraj, P. Thirathipviwat, J. Han, A. Gebert, Microstructure, mechanical and thermal oxidation behavior of AlNbTiZr high entropy alloy, Intermetallics 100 (2018) 9–19, https://doi.org/10.1016/j.intermet.2018.05.015.
[31] P. Zhang, Y. Li, Z. Chen, J. Zhang, B. Shen, Oxidation response of a vacuum arc melted NbZrTiCrAl refractory high entropy alloy at 800–1200 °C, Vacuum 162 (2019) 20–27, https://doi.org/10.1016/j.vacuum.2019.01.026.
[32] O.A. Waseem, H.J. Ryu, Combinatorial synthesis and analysis of $Al_xTa_yV_z$-$Cr_{20}Mo_{20}Nb_{20}Ti_{20}Zr_{10}$ and $Al_{10}CrMo_xNbTiZr_{10}$ refractory high-entropy alloys: Oxidation behavior, J. Alloy. Compd. (2020), 154427, https://doi.org/10.1016/j.jallcom.2020.154427.
[33] R. Gawel, Ł. Rogal, K. Przybylski, Oxidation resistance of Ti-Al-Cr-Nb-based high-entropy alloys in air at 1073 K, J. Mater. Eng. Perform. 287 (28) (2019) 4163–4170, https://doi.org/10.1007/S11665-019-04177-X, 2019.
[34] O.A. Waseem, U. Auyeskhan, H.M. Lee, H.J. Ryu, A combinatorial approach for the synthesis and analysis of $Al_xCr_yMo_zNbTiZr$ high-entropy alloys: oxidation behavior, J. Mater. Res. (2018) 1–9, https://doi.org/10.1557/jmr.2018.241.
[35] M.P. Brady, J.L. Smialek, J. Smith, D.L. Humphrey, The role of Cr in promoting protective alumina scale formation by γ-based Ti-Al-Cr alloys—I. Compatibility with alumina and oxidation behavior in oxygen, Acta Mater. 45 (1997) 2357–2369, https://doi.org/10.1016/S1359-6454(96)00362-X.
[36] M.P. Brady, J.L. Smialek, D.L. Humphrey, J. Smith, The role of Cr in promoting protective alumina scale formation by γ-based Ti-Al-Cr alloys— II. Oxidation behavior in air, Acta Mater. 45 (1997) 2371–2382, https://doi.org/10.1016/S1359-6454(96)00361-8.

[37] Z. Tang, V. Shemet, L. Niewolak, L. Singheiser, W.J. Quadakkers, Effect of Cr addition on oxidation behavior of Ti–48Al–2Ag alloys, Intermetallics 11 (2003) 1–8, https://doi.org/10.1016/S0966-9795(02)00062-6.
[38] F. Stein, A. Leineweber, Laves phases: a review of their functional and structural applications and an improved fundamental understanding of stability and properties, J. Mater. Sci. (2020) 1–107, https://doi.org/10.1007/s10853-020-05509-2.
[39] M. Zhu, L. Yao, Y. Liu, M. Zhang, K. Li, Z. Jian, Microstructure evolution and mechanical properties of a novel $CrNbTiZrAl_x$ ($0.25 \leq x \leq 1.25$) eutectic refractory high-entropy alloy, Mater. Lett. 272 (2020), 127869, https://doi.org/10.1016/j.matlet.2020.127869.
[40] F. Birks, N. Meier, G. Pettit, Introduction to the High Temperature Oxidation of Metals, second ed., Cambridge University Press, 2006 https://doi.org/10.1017/CBO9781139163903.006.
[41] N. Yurchenko, E. Panina, D. Shaysultanov, S. Zherebtsov, N. Stepanov, Refractory high entropy alloy with ductile intermetallic B2 matrix / hard bcc particles and exceptional strain hardening capacity, Materialia 20 (2021), 101225, https://doi.org/10.1016/J.MTLA.2021.101225.
[42] N. Yurchenko, E. Panina, Rogal, L. Shekhawat, S. Zherebtsov, N. Stepanov, Unique precipitations in a novel refractory Nb-Mo-Ti-Co high-entropy superalloy, Mater. Res. Lett. 10 (2022) 78–87, https://doi.org/10.1080/21663831.2021.2022033.
[43] N. Yurchenko, E. Panina, A. Belyakov, G. Salishchev, S. Zherebtsov, N. Stepanov, On the yield stress anomaly in a B2-ordered refractory $AlNbTiVZr_{0.25}$ high-entropy alloy, Mater. Lett. 311 (2022), 131584, https://doi.org/10.1016/J.MATLET.2021.131584.
[44] W. Lengauer, Properties of bulk δ-TiN_{1-x} prepared by nitrogen diffusion into titanium metal, J. Alloy. Compd. 186 (1992) 293–307, https://doi.org/10.1016/0925-8388(92)90016-3.
[45] α-Al2O3 (Al2O3) Crystal Structure - SpringerMaterials, (n.d.). ⟨https://materials.springer.com/isp/crystallographic/docs/sd_0315064⟩ (Accessed 18 April 2022).
[46] L. Ben-Dor, Y. Shimony, Structure and physical properties of the system $Nb_xCr_{1-x}O_2$, J. Magn. Magn. Mater. 7 (1978) 60–62, https://doi.org/10.1016/0304-8853(78)90150-6.
[47] M.A. Tena, P. Escribano, G. Monrós, J. Carda, J. Alarcón, Influence of niobate structures on the formation of rutile solid solutions [$M^{III}NbO_4$-1bTiO_2, where MIII=Al,Fe,Cr], Mater. Res. Bull. 27 (1992) 1301–1308, https://doi.org/10.1016/0025-5408(92)90095-H.
[48] TiO2 Crystal Structure - SpringerMaterials, (n.d.). ⟨https://materials.springer.com/isp/crystallographic/docs/sd_0310494⟩ (accessed April 18, 2022).
[49] S.B. Qadri, C.M. Gilmore, C. Quinn, E.F. Skelton, C.R. Gossett, Phase stability of $ZrO_2 - Al_2O_3$ thin films deposited by magnetron sputtering, Phys. Rev. B. 39 (1989) 6234, https://doi.org/10.1103/PhysRevB.39.6234.
[50] S. Sheikh, M.K. Bijaksana, A. Motallebzadeh, S. Shafeie, A. Lozinko, L. Gan, T.-K. Tsao, U. Klement, D. Canadinc, H. Murakami, S. Guo, Accelerated oxidation in ductile refractory high-entropy alloys, Intermetallics 97 (2018) 58–66, https://doi.org/10.1016/J.INTERMET.2018.04.001.
[51] C.E. Lowell, D.L. Deadmore, The role of thermal shock in cyclic oxidation, Oxid. Met. 144 (14) (1980) 325–336, https://doi.org/10.1007/BF00603788.
[52] C. Leyens, D.L.R. German, Oxidation and Protection of Titanium Alloys and Titanium Aluminides, 2003.
[53] P. Kofstad, Oxidation of Metals: Determination of Activation Energies, Nat 179 (1957) 1362–1363, https://doi.org/10.1038/1791362a0, 1957 1794574.
[54] C.J. Rosa, Oxygen diffusion in alpha and beta titanium in the temperature range of 932° to 1142°C, 1970, Metall. Trans. 19 (1) (1970) 2517–2522, https://doi.org/10.1007/BF03038377.
[55] J. Unnam, R.N. Shenoy, R.K. Clark, Oxidation of commercial purity titanium, Oxid. Met. 263 (26) (1986) 231–252, https://doi.org/10.1007/BF00659186, 1986.
[56] D.A. Venkatu, L.E. Poteat, Diffusion of titanium of single crystal rutile, Mater. Sci. Eng. 5 (1970) 258–262, https://doi.org/10.1016/0025-5416(70)90014-5.
[57] D. Huang, J. Lu, Y. Zhuang, C. Tian, Y. Li, The role of Nb on the high temperature oxidation behavior of $CoCrFeMnNb_xNi$ high-entropy alloys, Corros. Sci. 158 (2019), 108088, https://doi.org/10.1016/J.CORSCI.2019.07.012.
[58] S. Seal, S.C. Kuiry, L.A. Bracho, Surface Chemistry of Oxide Scale on IN-738LC Superalloy: effect of Long-Term Exposure in Air at 1173 K, 2002, Oxid. Met. 573 (57) (2002) 297–322, https://doi.org/10.1023/A:1014878402960.
[59] T.M. Butler, O.N. Senkov, T.I. Daboiku, M.A. Velez, H.E. Schroader, L.G. Ware, M. S. Titus, Oxidation behaviors of CrNb, CrNbTi, and CrNbTaTi concentrated refractory alloys, Intermetallics 140 (2022), 107374, https://doi.org/10.1016/J.INTERMET.2021.107374.
[60] Z. Zhang, M. Cheng, Z. Lu, Z. Yu, S. Liu, R. Liang, Y. Liu, J. Shi, R. Xiong, Magnetic properties and thermal stability of Ti-doped CrO_2 films, J. Magn. Magn. Mater. 451 (2018) 572–576, https://doi.org/10.1016/J.JMMM.2017.11.106.
[61] F. Müller, B. Gorr, H.J. Christ, J. Müller, B. Butz, H. Chen, A. Kauffmann, M. Heilmaier, On the oxidation mechanism of refractory high entropy alloys, Corros. Sci. 159 (2019), 108161, https://doi.org/10.1016/j.corsci.2019.108161.
[62] S. Schellert, B. Gorr, S. Laube, A. Kauffmann, M. Heilmaier, H.J. Christ, Oxidation mechanism of refractory high entropy alloys Ta-Mo-Cr-Ti-Al with varying Ta content, Corros. Sci. 192 (2021), 109861, https://doi.org/10.1016/J.CORSCI.2021.109861.
[63] S. Zeng, P. Li, J. Tian, C. Chen, Y. Meng, C. Zhu, H. Shen, X. Han, H. Zhang, Influence of Al content on the oxidation behavior of CrAl coating on Zry-4 alloys in 1200°C steam, Corros. Sci. (2022), 110115, https://doi.org/10.1016/J.CORSCI.2022.110115.
[64] Y. Du, D. Ding, L. Lai, S. Xiao, N. Guo, B. Song, S. Guo, Effect of Y on the high-temperature oxidation behavior of CrMoTaTi refractory high entropy alloy, Int. J. Refract. Met. Hard Mater. 103 (2022), 105755, https://doi.org/10.1016/J.IJRMHM.2021.105755.